

Notes: Brander and Lewis (AER, 1986)

Oligopoly and Financial Structure: The Limited Liability Effect

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model environment and key objects	4
3.1	Timing of the game	4
3.2	Operating profits and uncertainty	4
3.3	Debt, bankruptcy cutoff, and residual claimants	5
3.4	Equity value	5
3.5	Debt value	5
4	Output-market equilibrium	6
4.1	Why leverage raises desired output in the normal case	6
4.2	Symmetric increases in debt	6
4.3	Unilateral increases in debt	6
4.4	Debt value maximization versus equity value maximization	7
4.5	Debt-controlled comparative statics	7
5	Selection of debt levels	7
5.1	Debt as a first-stage commitment variable	7
5.2	Two effects of increasing debt	8
5.3	Positive debt in the normal case	8
5.4	Joint value and procompetitive debt	8
6	Identification and interpretation	9
6.1	What the model identifies	9
6.2	Why the sign of R_{iz}^i matters	9
6.3	Why fixed capital is assumed	9
6.4	Role of assumptions about debt contracts	9
6.5	Relation to Modigliani–Miller	10
6.6	Limitations	10

7	Figures	10
7.1	Figures 1 and 2: State-space division and reaction-function shifts	10
8	Short summary	11
9	Conclusion	12
9.1	Core conclusion	12
9.2	Economic intuition	12
9.3	Incremental contribution revisited	12
9.4	Implications	12
9.5	Limitations	13
9.6	Takeaway	13

1 Big picture

- **Research question.** Can a firm's financial structure affect its behavior in the product market? More specifically, can debt make an oligopolistic firm choose a more aggressive output policy, and can this strategic output effect help explain the debt-equity ratio?
- **Core idea.** With limited liability, shareholders receive the residual claim only in states where operating profits are high enough to cover debt. In bankrupt states, bondholders become residual claimants. Debt therefore changes which states shareholders care about when choosing output.
- **Limited-liability effect.** If marginal profits from extra output are higher in good states of the world, then debt removes low-marginal-profit states from the shareholders' objective. Equity holders then prefer a higher output level.
- **Strategic commitment.** In a Cournot oligopoly, higher debt can shift a firm's reaction function outward. A leveraged firm credibly commits to producing more aggressively, which induces its rival to reduce output.
- **Main theoretical result.** In the standard case $R_{iz}^i > 0$, common increases in debt raise symmetric equilibrium output, and a unilateral increase in firm i 's debt raises firm i 's output while lowering firm j 's output.
- **Debt choice.** When firms choose debt before choosing output, they recognize the strategic output benefit of debt. In the baseline model, firms choose positive debt when $R_{iz}^i > 0$ even though debt worsens the conflict between shareholders and bondholders.
- **Important reversal.** If $R_{iz}^i < 0$, so that marginal profits are lower in good states and higher in bad states, the signs reverse. Debt can make the firm less aggressive, and in the simplified debt-choice problem firms choose all equity.
- **Economic takeaway.** Financial structure is not just a way to divide an exogenous cash-flow stream. In oligopoly, debt can change the real cash-flow stream itself by changing product-market strategy.

2 Incremental contribution of the paper

- **Links corporate finance and industrial organization.** Standard capital-structure models usually treat product-market payoffs as exogenous. Standard oligopoly models usually ignore debt obligations. Brander and Lewis place the two decisions in the same game.
- **Debt as a product-market commitment device.** The paper adds financial structure to the set of strategic commitment variables in oligopoly theory, alongside capacity, location, R&D, and product choice.
- **New determinant of leverage.** The paper shows that output-market considerations can influence the optimal debt-equity ratio, even when taxes and direct bankruptcy costs are deliberately abstracted away.
- **Limited liability rather than only tax shields.** The mechanism is not a tax advantage of debt. Debt matters because limited liability changes the shareholders' payoff region and therefore their preferred output choice.

- **Agency insight with endogenous real decisions.** Jensen–Meckling and Myers emphasize shareholder–bondholder conflicts. Brander and Lewis show that the same conflict can matter for real product-market behavior when output is chosen after debt.
- **Strategic externality of leverage.** A firm’s debt affects not only its own objective but also its rival’s equilibrium output. This strategic externality means individually optimal debt choices need not maximize joint industry value.
- **Foundation for later product-market finance.** The paper helps launch a large literature on interactions among leverage, entry, pricing, output, predation, and stakeholder relations.

3 Model environment and key objects

3.1 Timing of the game

The paper studies a two-stage sequential duopoly game.

- **Stage 1: financial structure.** Firms choose debt obligations D_1 and D_2 .
- **Stage 2: output.** Firms choose output quantities q_1 and q_2 , taking debt levels as predetermined.
- **After output is chosen: uncertainty is realized.** A random state variable z_i affects firm i ’s operating profits.
- **Payoff settlement.** If operating profits cover debt, debtholders are paid in full and shareholders receive the residual. If not, the firm is bankrupt and debtholders receive the operating profits.

The equilibrium concept is sequential rationality: second-stage output choices form a Cournot equilibrium given debt levels, and first-stage debt choices anticipate that output equilibrium.

3.2 Operating profits and uncertainty

Firm i ’s operating profit is

$$R^i(q_i, q_j, z_i),$$

where q_i is own output, q_j is rival output, and z_i is a random state. Higher z_i is a better state for operating profits.

The model assumes standard Cournot-type conditions:

- own marginal profit falls with own output: $R_{ii}^i < 0$;
- rival output reduces firm i ’s profit: $R_j^i < 0$;
- better states raise operating profits: $R_z^i > 0$.

The crucial sign is the cross-partial R_{iz}^i , the effect of the state on marginal profit from output:

- $R_{iz}^i > 0$: marginal profits are higher in good states. This is the paper’s normal case.
- $R_{iz}^i < 0$: marginal profits are lower in good states. The main comparative statics reverse.

Interpretation. If $R_{iz}^i > 0$, extra output is like a risky action: it pays off especially well in good states and is costly in bad states. Debt makes shareholders more willing to take that action because they do not bear the full downside in bankrupt states.

3.3 Debt, bankruptcy cutoff, and residual claimants

The firm's debt obligation is D_i . Define the bankruptcy cutoff $\hat{z}_i$ by

$$R^i(q_i, q_j, \hat{z}_i) - D_i = 0.$$

At $z_i = \hat{z}_i$, the firm can just pay its debt. For $z_i < \hat{z}_i$, the firm is bankrupt and bondholders are residual claimants. For $z_i > \hat{z}_i$, shareholders are residual claimants.

The key derivative is

$$\frac{\partial \hat{z}_i}{\partial D_i} = \frac{1}{R_z^i(\hat{z}_i)} > 0.$$

Thus, higher debt raises the bankruptcy cutoff. More states become debt-relevant, and fewer states remain equity-relevant.

3.4 Equity value

When managers maximize shareholder value in the output stage, the relevant objective is expected equity value:

$$V^i(q_i, q_j; D_i) = \int_{\hat{z}_i}^{\bar{z}} [R^i(q_i, q_j, z_i) - D_i] f(z_i) dz_i.$$

The lower limit $\hat{z}_i$ is endogenous. A rise in debt shifts this lower limit upward, so shareholders ignore a larger set of low-profit states.

The first-order condition for output under equity value maximization is

$$V_i^i = \int_{\hat{z}_i}^{\bar{z}} R_i^i(q_i, q_j, z_i) f(z_i) dz_i = 0.$$

Interpretation. Shareholders choose output by averaging marginal profits only over the solvent states. Debt changes this average by removing the lowest states from the shareholder objective.

3.5 Debt value

For comparison, the paper also considers what output would be chosen if debtholders controlled the firm. The value of debt is

$$W^i(q_i, q_j; D_i) = \int_{\underline{z}}^{\hat{z}_i} R^i(q_i, q_j, z_i) f(z_i) dz_i + D_i [1 - F(\hat{z}_i)].$$

The first term is what debtholders receive in bankrupt states; the second term is what they receive in nonbankrupt states, where they are paid in full.

Interpretation. Equity holders and debtholders care about different parts of the state space. That is the source of their different preferred output policies.

4 Output-market equilibrium

4.1 Why leverage raises desired output in the normal case

Suppose $R_{iz}^i > 0$. Then marginal profits from extra output are lowest in bad states and highest in good states. Debt raises $\hat{z}_i$, so some low-marginal-profit states are removed from shareholders' decision calculus. The average marginal profit over the remaining equity-relevant states rises. To restore the first-order condition, the firm increases output.

This is the central limited-liability effect:

- debt expands the bankruptcy region;
- shareholders no longer care about marginal losses in newly bankrupt states;
- if those states have low marginal returns, the shareholder-optimal output rises;
- in Cournot competition, this aggressive stance affects rivals' choices.

A useful analogy is a team behind late in a game. Bad outcomes are already bad, so the team takes aggressive actions that raise upside payoff even if they worsen downside payoff. In the model, shareholders of a leveraged firm behave similarly.

4.2 Symmetric increases in debt

In a symmetric duopoly with $D_1 = D_2 = D$, the Cournot equilibrium is symmetric: $q_1 = q_2 = q$. Proposition 1 states:

- if $R_{iz}^i > 0$, equilibrium output q is increasing in the common debt level D ;
- if $R_{iz}^i < 0$, equilibrium output q is decreasing in D .

The implication is that a leveraged industry produces more than a fully equity-financed industry in the standard case. If $R_{iz}^i = 0$, financial structure has no output-market effect.

Economic message. Financial structure affects real decisions except in the special case where the marginal return to output is independent of the state of the world.

4.3 Unilateral increases in debt

Proposition 2 studies what happens when only firm i increases its debt.

- If $R_{iz}^i > 0$, firm i 's reaction function shifts outward.
- Firm i produces more for any given rival output.
- In equilibrium, firm i 's output rises and firm j 's output falls.
- If $R_{iz}^i < 0$, the signs reverse: firm i becomes less aggressive and firm j expands.

Economic message. Debt can be used strategically. A firm can make itself a tougher Cournot competitor by committing to a capital structure that induces aggressive output behavior.

4.4 Debt value maximization versus equity value maximization

The paper compares the shareholder-controlled firm with a hypothetical debtholder-controlled firm.

When $R_{iz}^i > 0$:

- shareholders prefer higher output because they are residual claimants in high states, where marginal profits are high;
- debtholders prefer lower output because they are residual claimants in low states, where marginal profits are low;
- therefore, equity-value-maximizing output exceeds debt-value-maximizing output.

This is Proposition 3. If $R_{iz}^i < 0$, the ranking reverses.

Interpretation. Debt creates a real agency conflict. Shareholders choose product-market policies that can reduce debt value, because shareholders enjoy the upside and pass part of the downside to debtholders.

4.5 Debt-controlled comparative statics

Proposition 4 shows that even if debtholders controlled the firm, the comparative statics of debt are qualitatively similar.

- In the normal case $R_{iz}^i > 0$, a symmetric increase in debt raises output even for a debt-controlled industry.
- A unilateral debt increase raises the debt-controlled firm's own output and lowers rival output.
- The intuition is that higher debt expands the range over which debtholders are residual claimants. When newly included states have higher marginal profit, debtholders also prefer more output.

Economic message. The direction of the debt-output effect is not simply about whether shareholders or debtholders are in control. It comes from how debt changes the relevant state space and how marginal returns vary across states.

5 Selection of debt levels

5.1 Debt as a first-stage commitment variable

The preceding analysis treats debt as given. The paper then asks how firms choose debt when they understand that debt will affect the later Cournot equilibrium.

Let

$$q_i = q_i(D_1, D_2)$$

denote the output chosen in the second-stage equilibrium. When firm owners choose debt in stage 1, they maximize total firm value, which is equity value plus debt value:

$$Y^i = V^i + W^i.$$

Since taxes and bankruptcy costs are absent in the baseline model, debt would be irrelevant if output were fixed. Debt matters only because it changes the later output equilibrium.

5.2 Two effects of increasing debt

A marginal increase in firm i 's debt has two opposing effects on total firm value.

- **Conflict-of-interest effect.** More debt worsens the divergence between shareholder-preferred output and total-value-maximizing output. This tends to reduce firm value.
- **Strategic output effect.** More debt changes rival behavior. In the normal case, firm i becomes more aggressive and firm j reduces output. Lower rival output raises firm i 's value.

At zero debt, the conflict-of-interest effect is zero because there are no debtholders. The strategic effect is positive when $R_{iz}^i > 0$. Therefore, a small amount of debt raises firm value in the normal case.

5.3 Positive debt in the normal case

Proposition 5 states:

- if $R_{iz}^i > 0$, equilibrium industry debt levels are strictly positive;
- if $R_{iz}^i < 0$, firms are entirely equity financed in the baseline model.

The result is deliberately stark because the model abstracts from taxes, bankruptcy costs, covenants, managerial agency problems, and signaling. The point is not that all leverage is explained by product-market strategy. The point is that product-market strategy is a distinct determinant of leverage.

Economic message. Firms may take on debt not because debt is a passive financing instrument, but because debt commits the firm to an aggressive output position.

5.4 Joint value and procompetitive debt

Proposition 6 shows that in the symmetric case, the debt levels chosen noncooperatively do not maximize joint industry value.

The reason is simple:

- standard Cournot oligopoly already produces too much output relative to the joint-profit-maximizing cartel outcome;
- when $R_{iz}^i > 0$, debt raises output even further;
- therefore, debt is procompetitive and reduces joint industry profits.

This is a prisoner's-dilemma logic. Each firm wants debt because it shifts the rival's output down. But when all firms use debt strategically, total output rises and joint profits fall.

6 Identification and interpretation

6.1 What the model identifies

This is a theoretical paper, not an empirical identification exercise. The paper identifies a logical channel through which limited liability can connect financial structure to product-market behavior.

The key comparative statics depend on:

- debt being chosen before output;
- output being chosen before uncertainty is fully resolved;
- limited liability shifting residual claims across states;
- marginal returns to output varying systematically with the state of the world;
- rivals observing or correctly anticipating the financial-structure choice.

6.2 Why the sign of R_{iz}^i matters

The cross-partial R_{iz}^i determines whether output is risk increasing or risk reducing from shareholders' perspective.

- If $R_{iz}^i > 0$, extra output pays off more in good states. Debt makes shareholders more willing to expand output.
- If $R_{iz}^i < 0$, extra output pays off more in bad states. Debt reduces shareholders' incentive to expand output.

The paper treats $R_{iz}^i > 0$ as the normal quantity-competition case, but it explicitly discusses alternatives such as advertising competition, where the reverse sign may arise.

6.3 Why fixed capital is assumed

The paper deliberately abstracts from physical investment. Capital stock is fixed before the debt-equity mix is chosen. Output changes come from variable inputs.

This assumption isolates the limited-liability effect. If investment were endogenous, debt could also affect output through capital choice, marginal cost, asset substitution, and salvage-value incentives. Those channels are important but separate from the mechanism the paper wants to identify.

6.4 Role of assumptions about debt contracts

The model assumes that output choices are not restricted by bond covenants or other precommitment devices. If creditors could write complete contracts governing later product-market behavior, the limited-liability effect could be muted.

The model also assumes that creditors correctly price debt, taking into account the future output equilibrium. Thus, the value of debt reflects bankruptcy risk. Debt is not a free transfer from naive bondholders.

6.5 Relation to Modigliani–Miller

In a Modigliani–Miller world, capital structure does not affect total firm value. Brander and Lewis obtain an exception because the firm’s real decisions are endogenous. Debt changes output choices, and output choices change the distribution of operating profits.

If output were fixed, and taxes and bankruptcy costs were absent, debt would again be neutral. The paper’s exception comes entirely from strategic interaction in the product market.

6.6 Limitations

- The paper is a stylized theory model, not a calibrated or estimated empirical model.
- It abstracts from taxes, direct bankruptcy costs, managerial agency problems, and debt covenants.
- The main result depends on the sign of R_{iz}^i and on Cournot-type strategic substitutes.
- The model assumes debt choices are observable and can serve as commitments.
- The paper focuses on limited liability; it explicitly leaves strategic bankruptcy effects to related work.

7 Figures

The original paper has two figures and no data tables. Both figures are reproduced below from the paper.

7.1 Figures 1 and 2: State-space division and reaction-function shifts

Read Figure 1:

- The horizontal line is the state space for z_i .
- Below the bankruptcy cutoff $\hat{z}_i$, debtholders are residual claimants.
- Above $\hat{z}_i$, equity holders are residual claimants.
- An increase in debt shifts the cutoff to $\hat{z}_i'$, expanding the debt-relevant region and shrinking the equity-relevant region.
- If $R_{iz}^i > 0$, the excluded states have low marginal returns to output, so shareholders choose a higher output level.

Read Figure 2:

- Figure 2 shows how a unilateral increase in firm 1’s debt shifts firm 1’s reaction function.
- In panel (a), $R_{iz}^i > 0$. Firm 1’s reaction function shifts outward, so firm 1 produces more and firm 2 produces less.
- In panel (b), $R_{iz}^i < 0$. Firm 1’s reaction function shifts inward, so firm 1 produces less and firm 2 produces more.

- This is the central strategic-commitment result: debt changes the firm's product-market stance.

Economic message: The diagrams show the entire mechanism in visual form. Debt changes which states matter to claimants, and that change translates into a shift in output reaction functions.

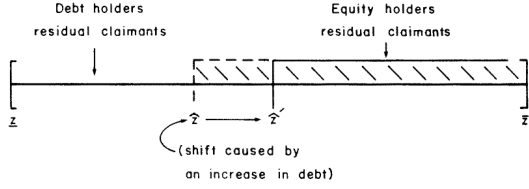
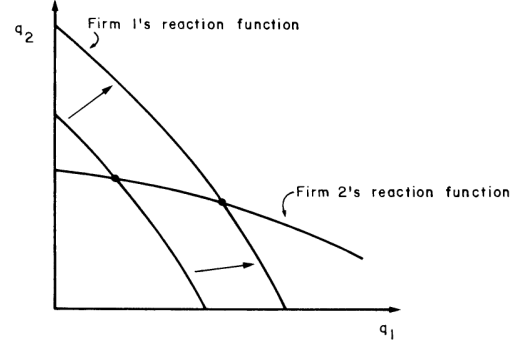
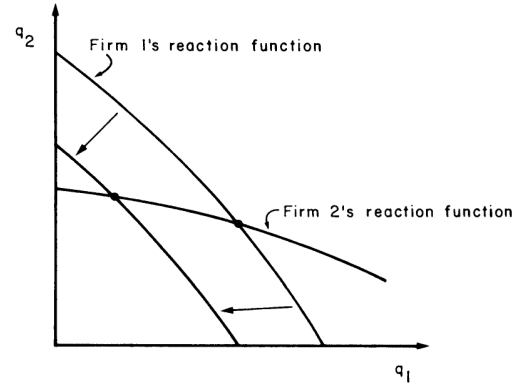


FIGURE 1. DIVISION OF STATE SPACE INTO DEBT-RELEVANT AND EQUITY-RELEVANT REGIONS

Figure 1: Division of state space into debt-relevant and equity-relevant regions.



(a) $R_{iz}^i > 0$



(b) $R_{iz}^i < 0$

FIGURE 2

Figure 2: Reaction-function shifts under $R_{iz}^i > 0$ and $R_{iz}^i < 0$.

8 Short summary

- The paper studies the interaction between financial structure and oligopoly behavior.
- Debt matters because limited liability changes the states over which shareholders are residual claimants.
- When marginal profits from output are higher in good states, debt makes shareholders prefer more output.
- In a Cournot duopoly, higher debt shifts a firm's reaction function outward.
- A unilateral increase in debt raises the leveraged firm's output and lowers rival output in the

standard case.

- Equity holders and debtholders prefer different output levels because they are residual claimants in different states.
- Firms choose positive debt in the baseline model because debt provides a strategic output advantage.
- Noncooperative debt choices reduce joint industry value because debt makes Cournot competition more aggressive.
- The model provides a new reason why capital structure can vary across industries: product-market strategy affects the value of debt as a commitment device.
- The paper's message is not that taxes, bankruptcy costs, and agency problems are irrelevant. Its message is that product-market interactions are an additional force in capital-structure choice.

9 Conclusion

9.1 Core conclusion

Brander and Lewis show that financial structure and product-market strategy cannot generally be analyzed separately. In an oligopoly, debt can change output decisions because limited liability changes the payoff states relevant to shareholders. In the normal case, leverage commits firms to a more aggressive output stance.

9.2 Economic intuition

Shareholders of a leveraged firm receive the upside in good states but are protected from part of the downside in bad states. If output expansion mainly increases payoff in good states, debt encourages expansion. Rivals anticipate this behavior, so leverage becomes a strategic commitment device.

9.3 Incremental contribution revisited

The paper's central contribution is to make capital structure endogenous to product-market rivalry. Debt is not merely a financial claim on a fixed cash-flow stream. It is a strategic variable that can alter the cash-flow stream by changing equilibrium output.

9.4 Implications

The model implies that industries with different modes of competition, demand uncertainty, cost uncertainty, and patterns of marginal profitability may have different optimal financial structures. It also implies that credit-market institutions and public policies affecting debt, such as interest deductibility, can have real product-market effects.

9.5 Limitations

The model is intentionally stripped down. It omits taxes, bankruptcy costs, managerial agency problems, covenants, and dynamic financing. These omissions make the mechanism transparent, but they also mean the model should be read as a theoretical channel rather than a complete theory of leverage.

9.6 Takeaway

The enduring takeaway is that leverage can be strategic. Under limited liability, debt can commit firms to aggressive product-market behavior, affect rivals' choices, and shape industry competition. Capital structure is therefore part of the real economy, not just a financial wrapper around it.